

FIN Legacy–Strict Bridge Next Proof Audit

Metadata

- Date UTC: `2026-07-19T19:00:58.050296+00:00`
- Git HEAD at generation: `b7d112f091aff320d26b722810dcfc1ccdbea8fd`
- Python: `3.14.4`
- NumPy: `2.5.1`

Executive verdict

This is a narrower proof-computational follow-up to the legacy-star audit. It constructs the explicit scalar/envelope/phase quotient objects between the reconstructed `K\*` legacy class and `K\_strict\_gate`, then separates finite algebraic witnesses from missing source theorems.

- **[Proven finite obstruction]** Positive scalar and positive damping/envelope maps cannot fix the `K\*` to strict sign mismatch; a phase/frequency transport object is mathematically necessary.
- **[Proven quotient fact]** `A` is a scalar multiplier: it drops out of normalized shape/sign comparisons, but it is still needed for unnormalized coupling strength, legacy `alpha\_geo` roles, moment magnitudes, and any future action-density normalization.
- **[Strong evidence]** `K\*` is easier to bridge than the rejected double-damped product because it is not exponentially dead and admits a clean positive damping quotient. It is still not close to strict without a sign-changing phase quotient.
- **[No-go on current artifacts]** The constructed quotient `Q(d)` is an algebraic finite assembly witness, not a source theorem, selector theorem, role-transfer theorem, unit theorem, or ToE closure.

Prior repo anchors used

```
| Anchor | Imported status |
| --- | --- |
| P2760 | seven open gaps: ontology-to-kernel measure, amplitude normalization,
phase/frequency/topological source, damping/compression bridge, moments-to-couplings provenance,
Lagrangian reverse closure, stale flags |
| P2691 | alpha_geo=4 ln 2 and scalar-shape normalization real; no role-safe amplitude
absorption source |
| P2692 | positive beta orbit/gauge representability real; no target-independent positive
beta/Z_beta source theorem |
| P3167_P3168 | no strict unit source S+ on current monomial/state map artifacts |
| scratch_bridge_assembly | existing finite assembly quotient maps legacy to strict on audited
nodes but explicitly does not export source, selector, role-transfer, or ToE closure |
```

Kernel definitions compared

```
K*_legacy(d;A,beta_l) = A*cos(pi*d/4 + pi/6)/(1+beta_l*d)
K_strict_gate(d) = cos(0.18575*d + 0.16250)/(1+d^1.8)
K_rej(d) = exp(-2.9*d)*(1+0.2*d)/(1+d)*cos(pi*d/4+pi/6)
```

Sign obstruction theorem

For every $A>0$ and every positive radial envelope $E(d)>0$, $\text{sign}(A \cdot E(d) \cdot \cos(\pi \cdot d/4 + \pi/6)) = \text{sign}(\cos(\pi \cdot d/4 + \pi/6))$; therefore positive amplitude/envelope completion cannot map the K^* sign row to the strict sign row.

```
| Row | Signs d=1..12 |
| --- | --- |
| strict | [1, 1, 1, 1, 1, 1, 1, -1, -1, -1, -1, -1] |
| K* | [1, -1, -1, -1, -1, 1, 1, 1, 1, -1, -1, -1] |
| K_rej | [1, -1, -1, -1, -1, 1, 1, 1, 1, -1, -1, -1] |
| K* mismatch distances | [2, 3, 4, 5, 8, 9] |
```

****Consequence:**** the bridge cannot be only `A` plus a positive damping/compression map. The missing theoretical object is a phase/frequency transport/source, not another scalar amplitude fit.

Damping/compression quotient object

For `beta\_l=0.05`, define `D\_beta(d)=(1+beta\_legacy\*d)/(1+d^1.8)`. On `d=1..12` it is positive: `True`, monotone decreasing: `True`, with `D(1)=0.525` and `D(12)=0.0180578`.

This is useful because it cleanly separates linear legacy damping from strict nonlinear compression after phase and amplitude have been factored away. It does **not** derive `eta=1.8`; it only says what positive envelope quotient would be needed if the phase/source layer were supplied.

Finite assembly quotient

After separating amplitude, phase, and damping, the exact finite quotient is:

$$Q(d) = A^{-1} * [\cos(\text{theta_strict}(d))/\cos(\text{theta_legacy}(d))] * [(1+\text{beta_l}*d)/(1+d^{1.8})]$$

$$K_legacy(d;A,\text{beta_l}) * Q(d) = K_strict_gate(d)$$

For `A=2.9`, `beta\_l=0.05`, the maximum absolute identity error on `d=1..12` is `5.551e-17`. This is an exact finite algebraic witness, but the phase factor ranges from `-1.78262` to `3.63177` and has `4` sign changes, so it is not a positive dissipative bridge.

What is `A`?

A is a scalar coupling/measure-density normalization multiplying the whole legacy profile; it is not a shape parameter.

Question	Answer
Does A drop from normalized shape?	True
If A=1, is it needed?	If one works only in quotient-normalized shape space, A=1 is a valid gauge. If one claims alpha_geo roles or absolute physical couplings, A cannot be deleted; it becomes an open source/normalization obligation.
A is needed for	['absolute unnormalized coupling strength', 'legacy role formulas that use alpha_geo explicitly', 'kernel moments before quotient/row normalization', 'any future action-density normalization theorem']
A is not needed for	['sign pattern', 'normalized profile shape', 'correlation/cosine similarity', 'row-stochastic Markov operator after full row normalization']

The value `A=4 ln 2=2.772588722239781` remains **[Proven]** as `ln 16`, the entropy of a uniform four-bit/16-state cell, but **[No-go]** as a physical unit, selector, or role-transfer source on current artifacts.

Is the new legacy kernel easier to bridge to strict?

Comparison	relative L2	cosine	scalar
unconstrained scalar K_rej -> strict	0.491662	0.870786	52.6998
positive scalar K_rej -> strict	0.491662	0.870786	52.6998
unconstrained scalar K*(beta=0.05) -> strict	0.997242	-0.0742191	-0.0208758
positive scalar K*(beta=0.05) -> strict	1	-0.0742191	0
best grid beta K* -> strict	0.997201	-0.0747643	beta=0.0741925, A_fit=-0.0233015

The scalar table is intentionally harsh: because the strict row is positive until d=7 while K* has negative entries at d=2..5, a positive scalar-only fit to K* collapses to zero. The rejected product can be inflated by a huge scalar to match mostly the first entry, but this is not a bridge; its norm is tiny and it is exponentially dead for d>=2. K* is therefore easier only after allowing the necessary phase/frequency object O_P plus the clean damping quotient O_D.

K* is easier than K_rej for bridge work because it preserves a non-dead long-range hyperbolic profile and separates damping from phase; it is not close to strict without a nontrivial phase/frequency map.

Constructed theoretical-object matrix

Object	Constructed here	Status	Missing for closure
--------	------------------	--------	---------------------

```

| --- | --- | --- | --- |
| O_A scalar amplitude/measure normalization | quotient factor  $A^{-1}$ ; proof that A drops from
normalized shape | witness/gauge decomposition only | source theorem fixing A as  $\alpha_{\text{geo}}$  or
proving amplitude absorption is role-safe |
| O_D damping/compression map |  $D_{\text{beta}}(d)=(1+\beta_l d)/(1+d^{1.8})$ , positive and decreasing on
 $d=1..12$  | finite/exact envelope witness after phase separation | dynamical source of  $\eta=1.8$  and
target-independent  $\beta/Z_{\text{beta}}$  theorem |
| O_P phase/frequency transport |  $P(d)=\cos(\theta_s)/\cos(\theta_l)$  finite quotient on  $d=1..12$  |
algebraic quotient with sign changes, not source law | phase/frequency/topological source
localizer and selector-safe transport theorem |
| O_S selector/orientation source | positive scalar/envelope sign no-go matrix | obstruction:
QW-2191 remains open | non-premise orientation/polarity source plus coupling-polarity theorem |
| O_U unit/action-density bridge | none; imported no-go status from
guardrails/P2760-P2764/P3167-P3168 | no-go on current artifacts | canonical
reference-cell/action-density theorem with physical units |

```

Recommended next honest step

****Recommendation:**** Do not run another generic bridge replay. Attack exactly O_P: construct or refute a phase/frequency transport source localizer for $P(d)=\cos(\theta_s)/\cos(\theta_l)$, while conditioning O_A and O_D as quotient witnesses only.

****Why:**** The new K* already repairs the historical envelope, so the remaining hard obstruction is not scalar A and not positive damping alone; it is the sign-changing phase/frequency/source layer plus the still-open unit/selector obligations.

****Success criterion:**** A machine-checkable theorem or finite exhaustive certificate that $P(d)$ comes from a strict exported source object and is selector-safe; otherwise emit a bounded no-go for this exact P object.

Do not promote this audit to strict bridge closure, selector closure, role transfer, physical units, ` $L_{\text{total}}$ `, or ToE.